

Closed-Loop Config in SONiC

An ACK proves the intent was recorded, not that the change was realized. Replace sleep timeouts with realization tracking.

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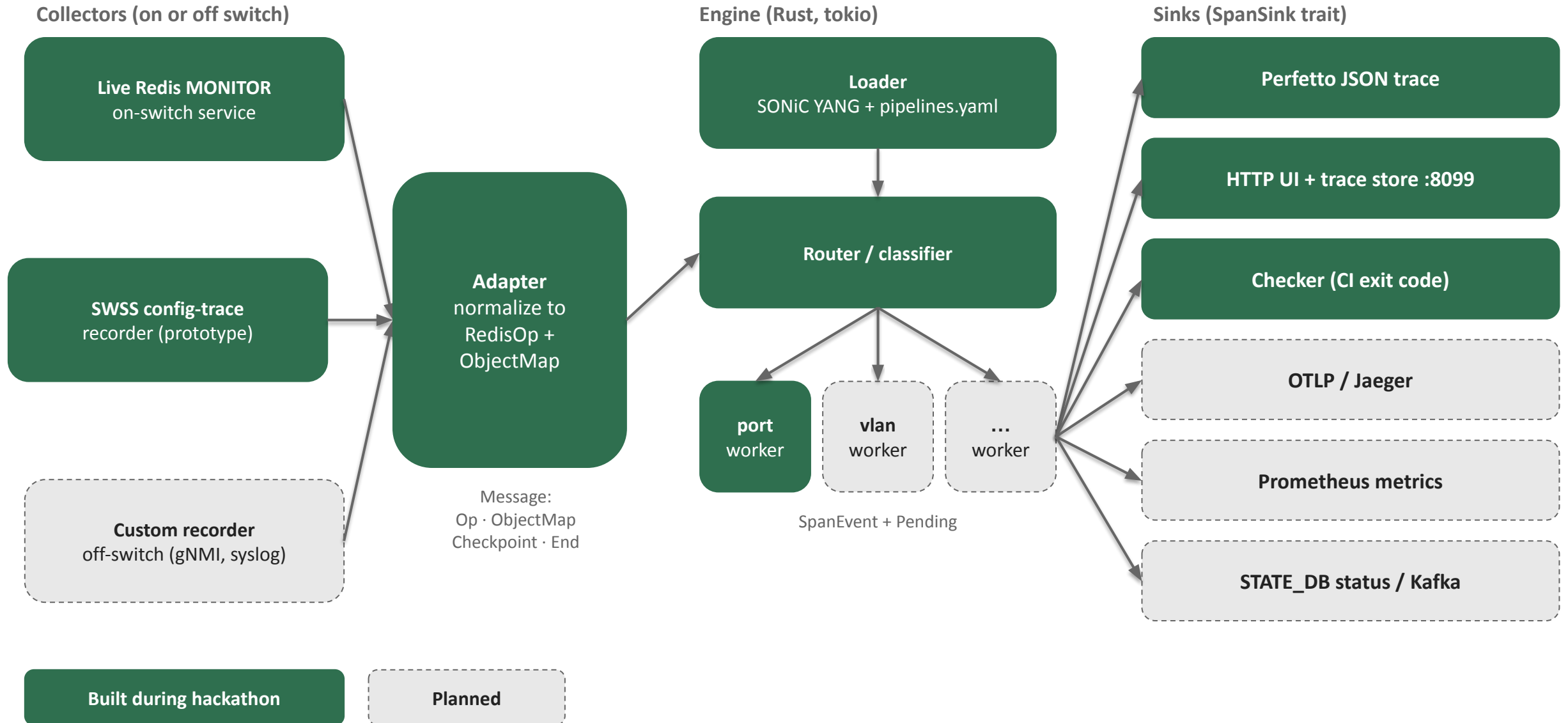
Category: CLI/config/tracking improvements

The problem: ACK is not realization

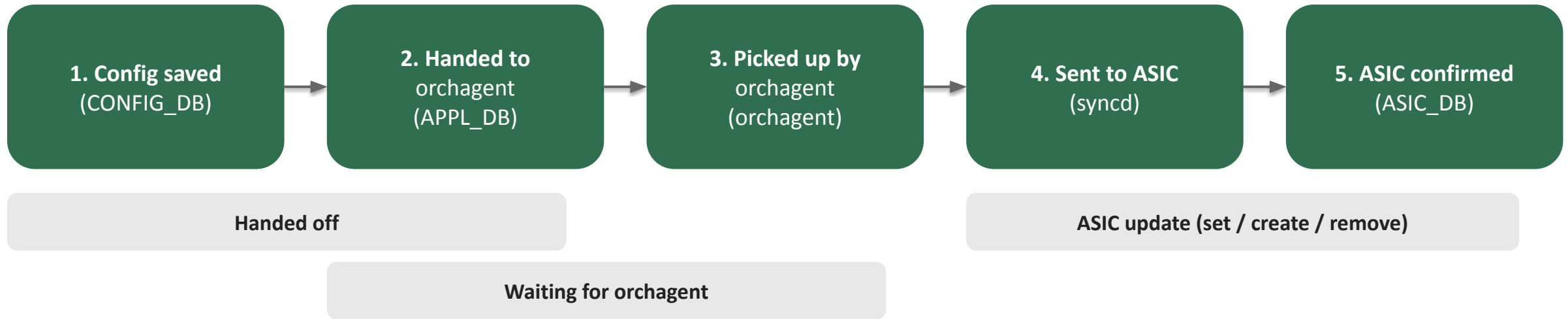
- CLI and gNMI requests are **acknowledged on intent validation**, before the change is processed
- The pipeline is **asynchronous and decoupled**: no inline feedback if a change **stalls, drops or fails**
- Automation relies on **arbitrary sleep timeouts** or **out-of-band state checks**: a barrier to zero-touch operations
- **Goal**: end-to-end realization tracking, **no changes to data models or schemas**



Architecture: realization tracking



How a change is tracked



- **Nothing dropped silently:** every setting you write must reach the next stage. If one doesn't, it's flagged.
- **One command, several hardware updates:** an MTU change updates both the port and its router interface. The change counts as done only when both finish.
- **Shows where it stopped:** not passed on by portmgrd · waiting for orchagent · sent to the ASIC, not confirmed · ASIC change with no matching config
- **No per-feature code:** what to track is read from SONiC's own YANG models.

Demo

Demo video

In Google Slides: select this box, delete it, then
Insert → Video → Google Drive → pick the demo video

What to watch for

- **Normal path:** CONFIG → APPL → orchagent → SAI PORT + RIF, with per-stage timings
- **orchagent paused:** CLI says OK, change stuck **waiting for orchagent**
- **syncd paused:** stuck one stage later, **waiting for the ASIC to confirm**
- **Checker:** non-zero exit while anything is pending

Backup: offline replay of the saved recording → open the trace in ui.perfetto.dev

Status and links

Repo: <https://github.com/karthik-aria/sonic-hackathon-2026>

Completed

- Port pipeline: CONFIG → APPL → SAI → ASIC_STATE
- Live service: Redis MONITOR + capture windows
- Web UI :8099 with Perfetto traces
- Docker image + systemd unit
- Offline replay (recording, config-trace)
- CI checker: non-zero exit if pending
- YANG-driven pipeline loader

Planned

- Status-tracking primitives in the messaging framework (per-change epoch stamps)
- Lightweight recording framework on the switch

Status: Partially completed